GRANT SELECTION DESIGN AND START-UP INNOVATION: EVIDENCE FROM DESIGN-DEPENDENT SUBSIDY EFFECTS

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Abstract

This project studies how public small and medium sized enterprises (SME) and start-up funding programmes use observable signals in the selection rules when allocating grants under imperfect information. This project examines whether these selection rules generate three outcome layers that motivate public support: technological innovation, observable spillover and economic value. Empirically, the project codes programme documents, determines implied signal weights, and estimates causal effects at thresholds. The contribution of this project shifts programme evaluation from beneficiary status to the economic content of the selection rules.

Research Objective and Research Question

The project studies whether and how public MSME funding programmes use observable signals in the selection process (funding decision) that allocate grants. It examines whether the chosen selection signals generate the three outcome layers that motivate public support: technological innovation, observable spillover and economic value.

The core research question is:

> *Do the selection signals used in public SME and start-up funding programmes generate technological innovation, observable spillover, and economic value — and do programmes that define value differently weight signals differently?*

The project examines three connected sub-questions:

1. How do public funding programmes translate policy objectives into selection rules — which selection signals do those rules weigh, and with which stated rationales?
2. Which outcomes do signal-based selection rules generate across the three outcome layers — technological innovation, observable spillover and economic value?
3. Do the signal weights used in the selection rules align with the objectives policy makers seek to achieve — and how large is the gap where they diverge?

This structure separates the selection mechanism from the grant. The project focuses on selection signals; grant features matter to the extent that they influence the selection rules.

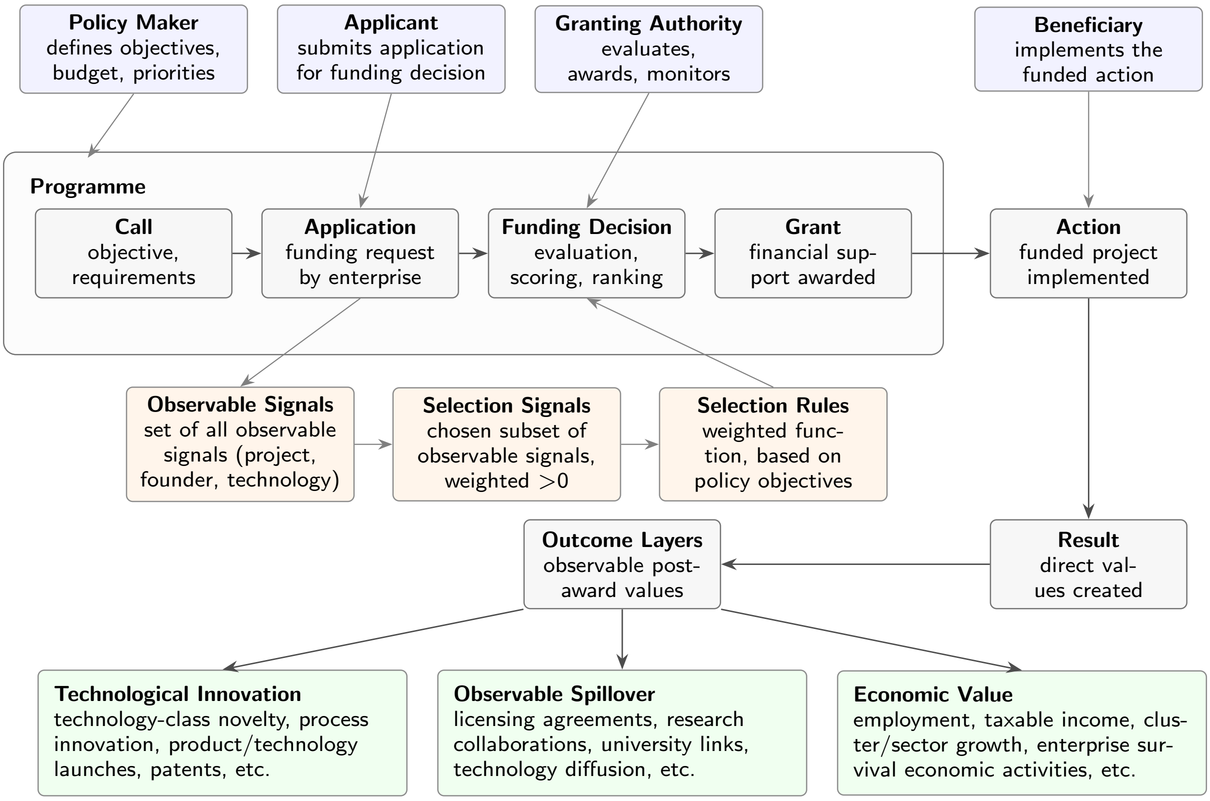
Motivation

Private returns to innovation fall systematically short of social returns (Arrow, 1962): new technologies generate knowledge, applications, and productivity effects that reach beyond the innovating enterprise. Bloom et al. (2013) show that R&D produces technology spillovers across enterprises and technology classes.

This is an appropriability problem which is sharpest for SUSME. Their projects are uncertain, often intangible and difficult to finance, precisely where the funding gap for R&D is widest (Hall and Lerner, 2010). Early-stage grants can close this gap: they increase patenting, revenue and follow-on private investment for financially constrained enterprises (Howell, 2017).

Grants do not resolve the appropriability problem. They mitigate it by reducing private funding requirements for the enterprise. Granting authorities select beneficiaries before outcomes are observed. At the time of the funding decision, only observable application signals are available: enterprise age, size, region, technology class, founder history, prior patents, research links etc. . These observable signals constitute the pool of available selection signals. The latter constitute the basis from which policy objectives translate into funding decisions, by applying the selection rules.

Which projects are funded depends on the selection rules. If calibrated inadequately, public support can displace private R&D rather than creating additional innovation (Wallsten, 2000). The decision to respond to a programme call, agency screening, enterprise actions and results are linked: the selection rule is intertwined with the grant effect (Takalo et al., 2013). Therefore, the central policy maker challenge is, whether the selection rules adequately identify projects, that create value beyond the beneficiaries. Innovation is understood broadly - as novel or improved products and processes made available to potential users (OECD and Eurostat, 2018).



Key Literature and Research Gap

The evaluation literature shows that public funding programmes increase technological innovation (Klette et al., 2000). Early-stage R&D grants increase patenting, patent quality, revenue and follow-on private investment; with effects concentrated among financially constrained enterprises (Howell, 2017). Regression-discontinuity designs around score and budget cut-offs provide credible identification of these effects (Bronzini and Iachini, 2014). Grants also alter the development path of young technology enterprises, with sustained employment growth and follow-on private investment (Lerner, 1999; Howell, 2017).

The same evidence reveals the central condition for public value: support must reach projects that would not have been financed otherwise. Where it does not, public R&D displaces private investment (Wallsten, 2000). Systematic reviews of the econometric evidence find no uniform answer; whether public R&D crowds in or out depends on programme design and targeting (David et al., 2000). Even where average effects are positive, adverse selection undermines full additionality: some funded projects would have proceeded without support, reducing net public investment additionality (Lach et al., 2021).

Effects not only depend on the existence of support, but on how support is allocated across enterprises, technologies and regions (Criscuolo et al., 2019).

If public support is justified by value beyond the beneficiaries' private gain, beneficiary growth is not a sufficient measure. R&D generates observable spillovers: social returns substantially exceed private returns (Bloom et al., 2013). Publicly funded R&D diffuses across enterprises, technology classes and geographic space (Myers and Lanahan, 2022). Selection rules that identify commercially promising enterprises may be well-designed for private value but poorly for public value. To this end the project tracks technological innovation, observable spillover and economic value as separate outcome layers: they constitute the aggregate the value that public support is meant to generate.

Two problems persist, independently of the quality of the funding decision: not all potentially eligible applicants apply for funding and the funding decision will always have some form of error. Applicants must actively apply. Adverse selection is not a correctable error but a structural deficiency of programmes that select under imperfect information (Lach et al., 2021). Therefore, application, screening and funding decisions must be studied jointly (Takalo et al., 2013).

The research gap is therefore a design gap. Prior work shows that patenting history and venture-capital backing are associated with both application propensity and funding likelihood in European SME programmes (Mina et al., 2021). Whether the selection signals used in selection rules align with policy makers stated objectives has not been examined. The evaluation literature treats these as controls, assignment variables, or institutional details (Klette et al., 2000), not as the primary object. We know grants can affect innovation and that public and private returns can diverge. We do not know whether the selection signals granting authorities use in selection rules generate the outcome that motivated public support. This project fills that gap.

Theoretical and Empirical Framework

The project treats funding programmes as a selection mechanism under imperfect information. The theoretical objects are the selection rules: granting authorities choose a subset of observable signals to define the selection signals, which form the basis of the funding decision. Selection rules typically combine absolute hurdles with a scoring function that ranks qualifying applicants within the available budget.

The empirical design has two steps. The first identifies the selection rule: which signals are used, with what weights, and for which stated objectives. The second estimates whether signal-based selection rules generate the forms of value the programme seeks to create. The unit of analysis is the enterprise-application pair.

The first step achieves two things. Programme call documents, scoring rubrics and eligibility criteria are coded to record which observable signals each programme states it judges and which policy objective each signal serves. The distribution of funded and unfunded applicants is subsequently analysed to recover implied scoring weights. Comparing stated weights with implied weights reveals where the defined selection rules diverge from the applied selection rules.

The outcome is split into three layers by who captures the value created. The first, technological innovation, spans from non-delivery – project cancellation or early termination – to high-quality outcomes. The second, observable spillover, captures value flowing from the beneficiary to others through knowledge transfer, licensing, collaboration, etc. . The third, economic value, captures value created in the region: employment, enterprise survival and sector growth. Outcome measures overlap across layers - a patent evidences technological innovation and observable spillover - so layer specific effects are estimated jointly.

The second step estimates the causal effect of selection on each outcome layer for applicants at the selection margin. A direct comparison of funded and unfunded enterprises would be misleading - they differ in ways that also predict their outcomes. The regression-discontinuity design (Imbens and Lemieux, 2008) resolves this: applicants just above and just below the score cut-off are otherwise similar. Therefore, differences in outcomes are caused by the funding decision itself. The key question is whether this causal effect differs by signal composition. That differential, not the average grant effect already established by Howell (2017) and Bronzini and Iachini (2014), is the primary causal quantity.

Where score data is available, the preferred design is regression-discontinuity around score or budget cut-offs. Where score data is unavailable, programme rule changes across calls, regions, or countries support panel event-study or Difference-in-Differences designs (Schmidheiny and Siegloch, 2023; Callaway and Sant'Anna, 2021). Expected threats are self-selection into application, score manipulation near cut-offs, survivorship bias and outcome correlation across layers.

The preferred data link programme records to innovation, enterprise and regional outcomes at the applicant level (e.g. from CORDA database/ORBIS). The first step draws on programme call documents, scoring rubrics, and beneficiary lists. EU state aid transparency requirements mandate disclosure of all beneficiaries, covering national and EU-level programmes. The second step links applicant-level award data to patent records from the European Patent Office (EPO/Patstat), enterprise accounts, business registers, and regional employment data. Score-level data from programmes such as the EIC Accelerator and the EU Transparency Register are a preferred starting point for the regression-discontinuity design.

Expected Outcome

The project will produce three connected findings. The first identifies which observable application signals granting authorities use in the selection rules and how these vary with stated programme objectives. The second establishes which outcomes the selection signals generate across the three outcome layers. The third assesses whether signal weights align with programme objectives and quantifies the gap where they diverge. Three falsifiable hypotheses follow from the theoretical framework and the outcome:

- Signal weights diverge from stated programme objectives over successive calls, reflecting accumulated pressure on evaluation practice rather than a change in signal choice.
- Selection signals that identify financial constraints generate higher additionality than signals that identify technological novelty. At the selection margin, enterprises funded on novelty signals may have attracted private funding; the enterprises funded on constraint signals would not have proceeded without support.
- Observable spillover depends not only on selection signal content but on the technological and geographic density of the target environment. The same selection rules generate more spillover in innovation-dense regions; signal reweighting alone cannot close the gap between innovation and spillover outcomes.

If the latter two are confirmed, the project will have identified a design gap: selection signals granting authorities use to select beneficiaries do not fully serve the policy objectives. This has direct implications for future design: if the selection rules cannot generate additionality or observable spillover through selection signals, the public case rests on narrower grounds than current programme objectives claim.

The project makes an empirical, conceptual, and design-oriented contribution. Empirically, it will assess whether selection signal weights generate technological innovation and additionality, and whether selection signals alone determine observable spillover and economic value. Conceptually, the project separates the effect of receiving grants from the economic contents of the selection rules based on which grants are allocated. This distinction matters: public innovation policy is not only about funding more projects, but about which pre-funding characteristics should guide scarce public support when outcomes are uncertain.

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